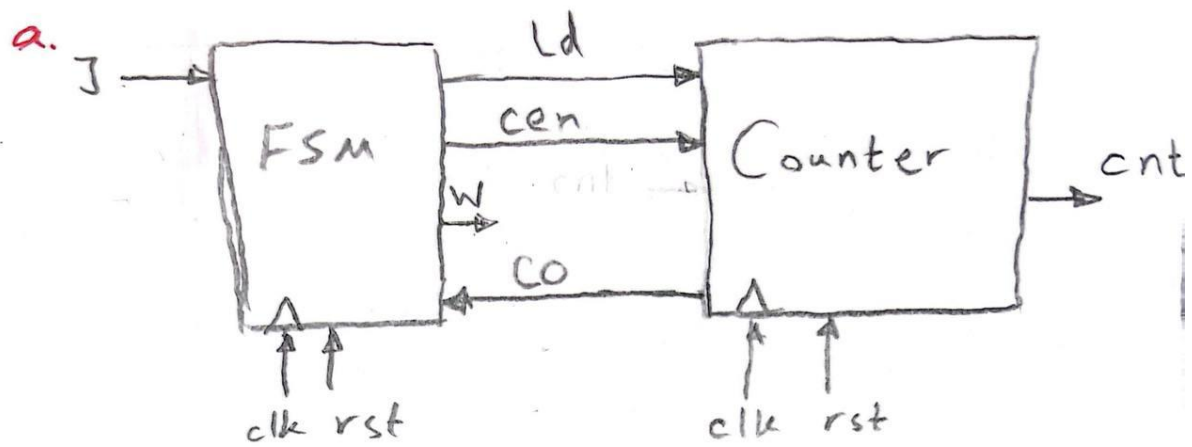
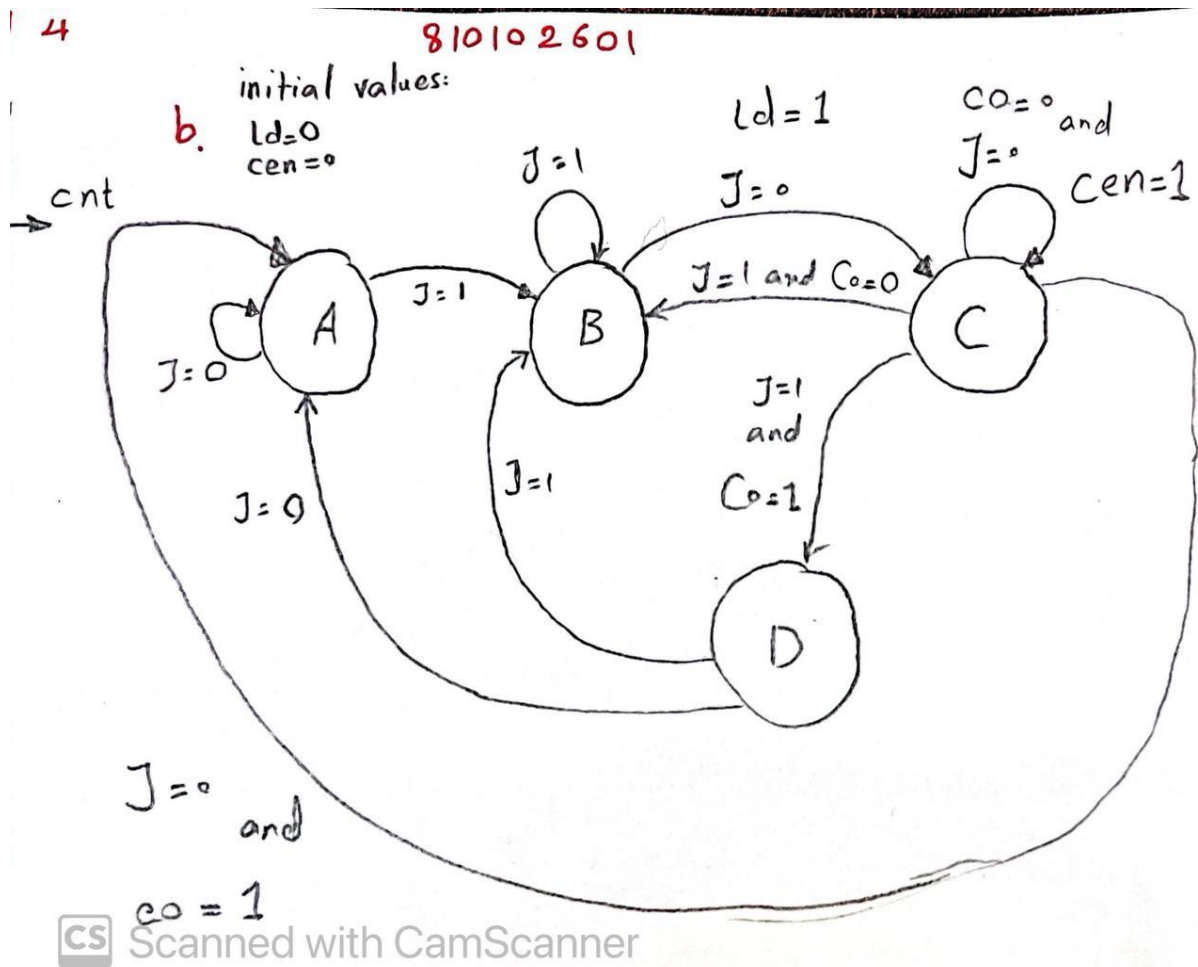


a.

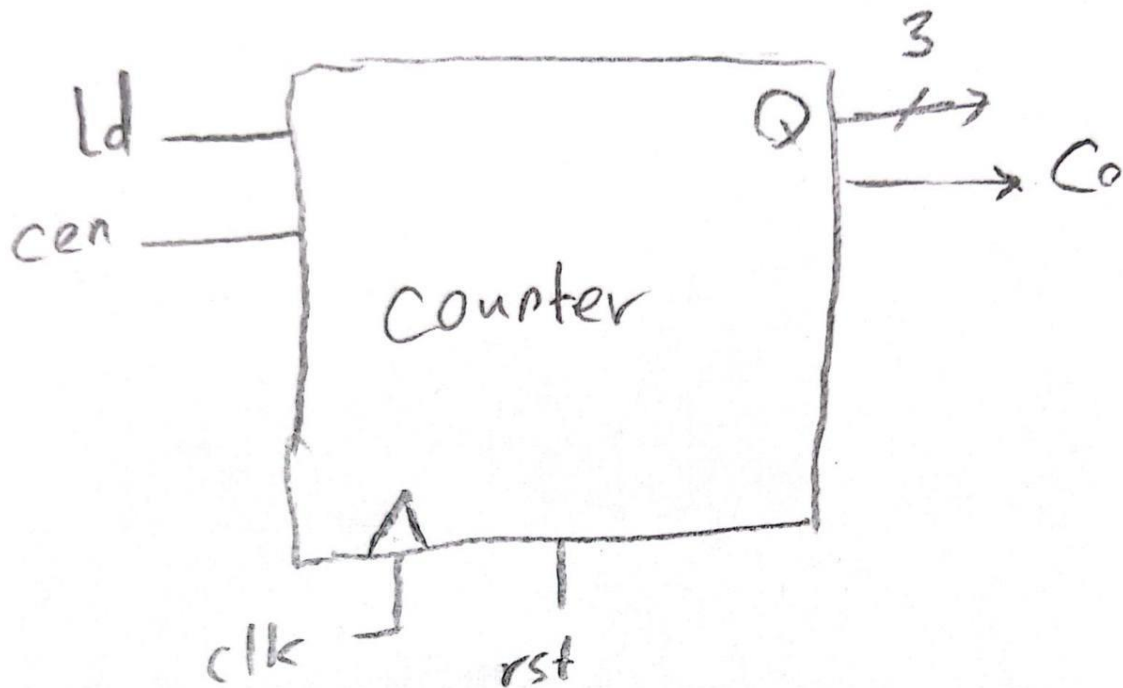


Scanned with CamScanner

b.



c.



CS Scanned with CamScanner

d)

```
`timescale 1ns/1ns
module partd1 (input clk, rst, j, output w);

    parameter A = 2'b00, B = 2'b01, C = 2'b10, D = 2'b11;
    logic [1:0] ps, ns;

    logic [2:0] cnt;
    logic cen, ld, co;

    always @(posedge clk or posedge rst) begin
        if (rst)
            ps <= A;
        else
            ps <= ns;
    end

    always @(ps , j , co) begin

        ld = 0;
        cen = 0;
        ns = ps;

        case (ps)
```

```

    A: ns = j ? B : A;

    B: begin
ld = 1;
        if (~j) begin
            ns = C;
        end else
            ns = B;
    end

    C: begin
        if (~j) begin
            if (~co) begin
                ns = C;
                cen = 1;
            end else
                ns = A;
        end else begin
            if (co)
                ns = D;
            else
                ns = B;
        end
    end

    D: ns = j ? B : A;

    default: ns = A;
endcase
end

always @(posedge clk or posedge rst) begin
    if (rst)
        cnt <= 3'b000;
    else begin
        if (ld)
            cnt <= 3'b001;
        else if (cen)
            cnt <= cnt + 1;
    end
end

always @(cnt) begin
    if (cnt == 3'b110)
        co = 1'b1;
    else
        co = 1'b0;
end

assign w = (ps == D) ? 1'b1 : 1'b0;

endmodule

```

e)

```
`timescale 1ns/1ns
```

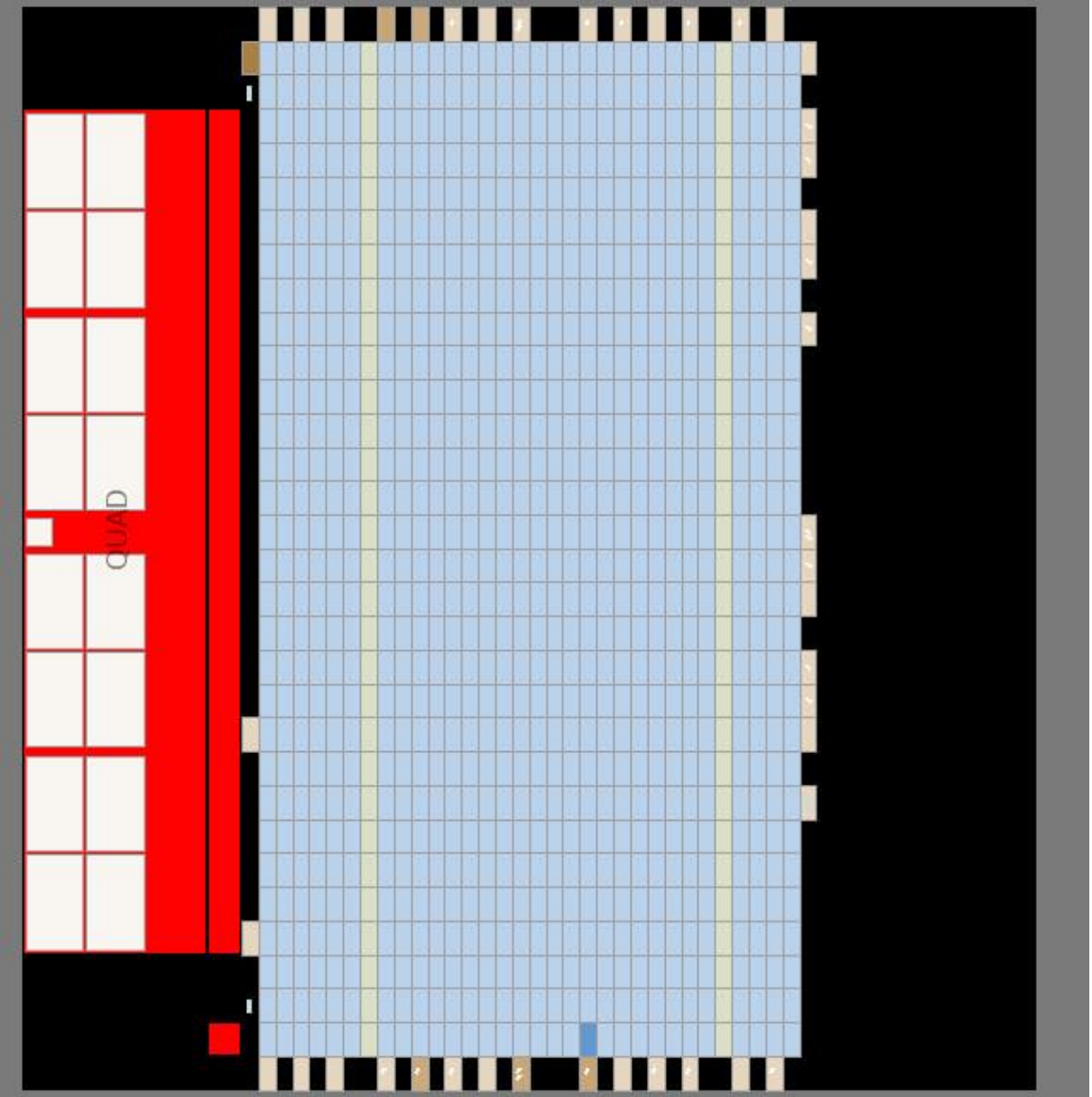
```

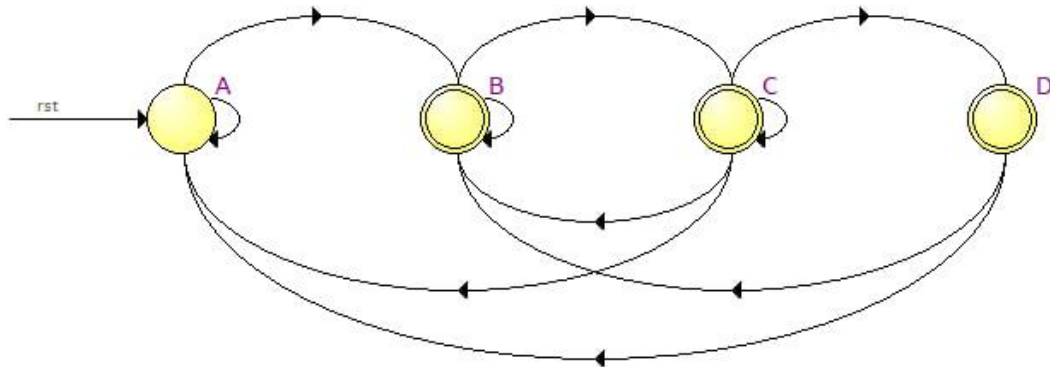
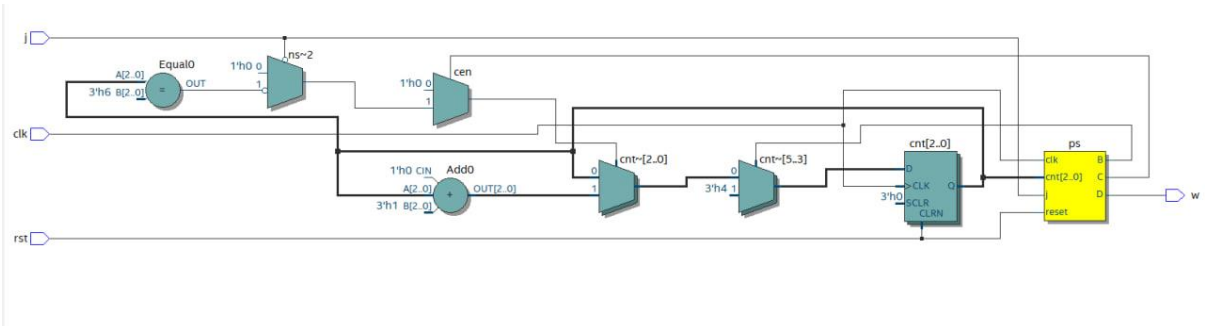
module parte();
    logic clk , rst , j ;
    logic w;
    // logic clk0 , rst0 , j0 ;
    logic w0;
    partd CUT(clk , rst , j , w);
    partd1 CUT0(clk , rst , j , w0);
    initial begin
        clk = 0;
        forever #10 clk = ~clk;
    end
    initial begin
        rst = 1;
        j = 0;
        #12 rst = 0;
        #1 j = 1;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 1;
        #20 j = 1;
        #20 j = 0;
        #20 j = 1;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 0;
        #20 j = 1;
        #20 j = 1;
        #20 j = 0;
        #40
        $stop;
    end
endmodule

```

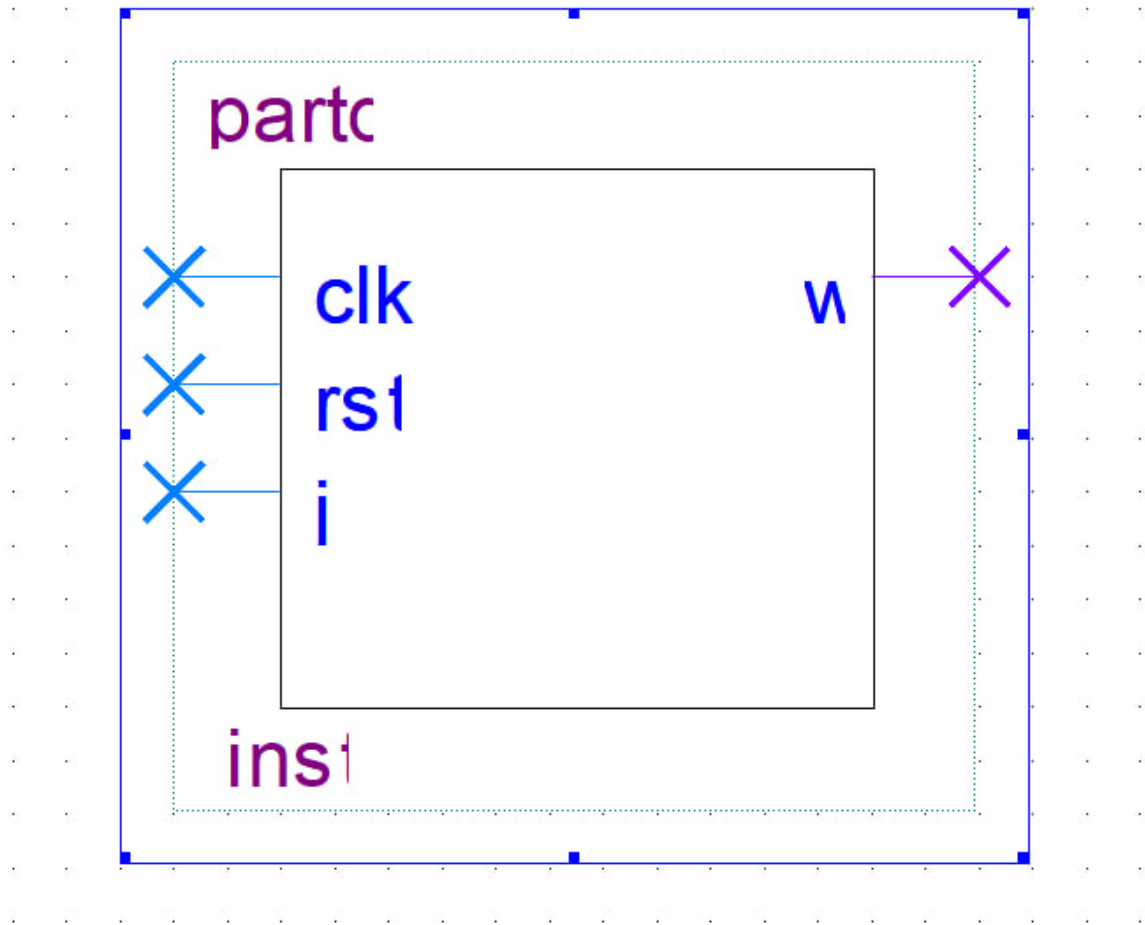
note: CUT0 .

f)





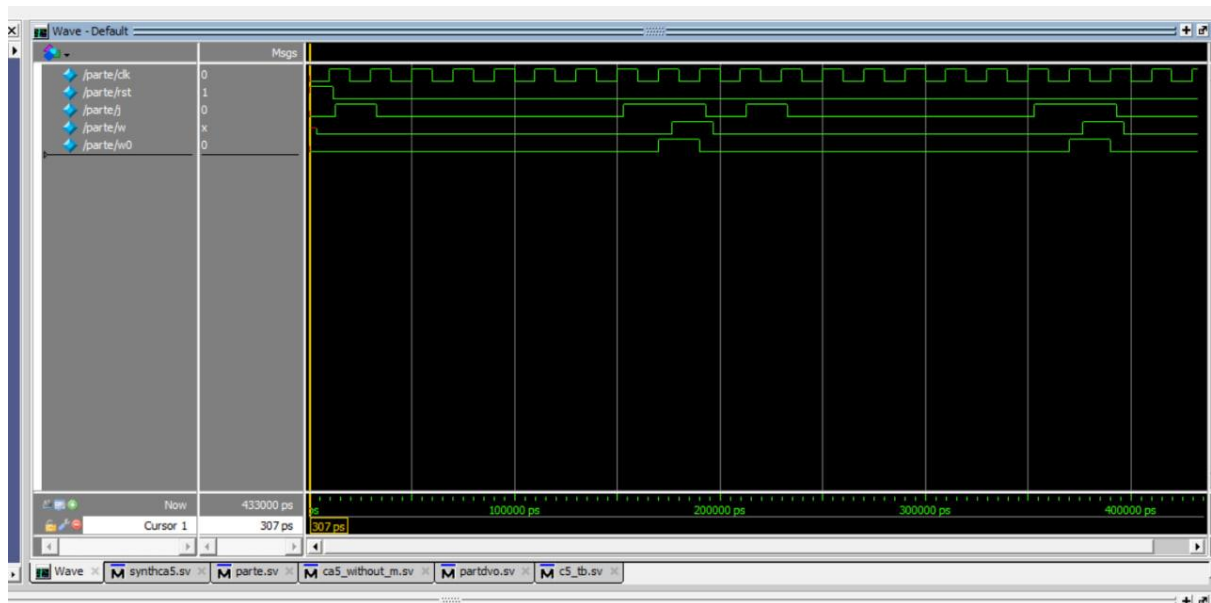
	Source State	Destination State	Condition
1	A	B	(j)
2	A	A	(!j)
3	B	C	(!j)
4	B	B	(j)
5	C	D	(!cnt[0]).(cnt[1]).(cnt[2]).(j)
6	C	C	(!cnt[0]).(!cnt[1]).(!j) + (!cnt[0]).(cnt[1]).(!cnt[2]).(!j) + (cnt[0]).(!j)
7	C	B	(j).(!cnt[0]).(!cnt[1]) + (j).(!cnt[0]).(cnt[1]).(!cnt[2]) + (j).(cnt[0])
8	C	A	(!j).(!cnt[0]).(cnt[1]).(cnt[2])
9	D	B	(j)
10	D	A	(!j)



g)

the test bench is same as part e

waveform image is bellow:



This testbench is used for both pre- and post-synthesis simulations, and as seen in the waveform, the delay becomes evident after synthesis.